

ST ANTHONY, NL, Canada

WMO: 718300

Lat: 51.390N

Lon: 56.090W

Elev: 108

StdP: 14.64

Time Zone: -3.50 (NAF)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.6	-4.9	-16.2	2.3	-7.3	-12.4	2.8	-4.3	41.8	20.7	37.9	19.6	13.2	270	0.886

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	73.6	62.5	71.0	61.7	68.3	60.3	65.2	70.6	63.3	68.6	61.5	66.8	18.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	86.5	68.0	61.0	80.3	66.2	59.0	74.9	64.5	30.2	70.9	28.8	68.8	27.5	66.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.8	31.1	28.1	-15.9	79.3	3.5	1.9	-18.4	80.7	-20.5	81.8	-22.4	82.9	-25.0	84.3		
			-16.3	69.0	3.5	1.5	-18.8	70.1	-20.9	71.0	-22.8	71.8	-25.3	72.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	14.6	14.4	21.1	30.5	39.6	49.0	57.6	58.5	51.1	41.0	31.5	21.8
	DBStd	17.04	10.50	10.37	9.30	5.85	5.36	6.53	6.03	5.28	6.37	5.89	7.25	9.57
	HDD50	5791	1096	996	895	586	328	98	10	3	61	287	556	875
	HDD65	10596	1561	1416	1360	1036	789	479	237	209	419	745	1005	1340
	CDD50	686	0	0	0	0	4	69	245	266	94	7	1	0
	CDD65	15	0	0	0	0	0	0	7	7	1	0	0	0
	CDH74	63	0	0	0	0	0	5	35	21	2	0	0	0
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	13.7	14.9	15.3	15.7	14.7	12.8	12.6	11.8	11.5	13.1	13.1	13.7	14.8
Precipitation	PrecAvg	43.3	4.4	3.6	3.1	2.4	2.7	3.2	3.9	3.7	3.7	3.5	4.1	
	PrecMax	51.8	13.5	5.9	5.6	5.0	6.2	7.0	7.9	8.5	8.5	6.4	5.7	6.5
	PrecMin	32.8	2.1	1.2	0.8	0.5	0.6	1.5	1.5	0.7	1.5	1.0	1.5	2.0
	PrecStd	5.0	2.5	1.3	1.1	1.1	1.5	1.4	1.5	1.6	1.4	1.3	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.9	38.8	39.0	48.6	63.3	73.2	77.8	76.5	72.0	60.7	55.9	42.8
		MCWB	39.2	37.0	36.5	42.0	53.0	60.0	64.7	64.3	63.6	54.3	53.0	41.8
	2%	DB	35.1	34.9	36.1	42.5	57.3	68.4	74.2	73.3	67.1	56.3	48.6	39.1
		MCWB	34.2	33.5	33.8	38.8	49.7	57.6	62.9	63.0	60.4	52.4	46.8	38.1
	5%	DB	32.6	32.6	34.3	39.8	53.1	64.7	71.4	70.6	64.2	53.1	44.1	36.3
		MCWB	31.7	31.4	32.6	36.7	46.8	55.6	62.1	62.0	58.5	49.8	42.4	35.4
	10%	DB	30.2	30.3	32.8	37.5	49.2	61.1	68.5	68.2	61.1	50.0	41.1	33.9
		MCWB	29.1	29.3	31.2	34.7	44.3	53.9	60.3	61.0	56.5	46.9	39.6	32.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	37.5	37.2	44.1	54.4	62.7	67.8	67.5	65.0	57.0	53.2	41.8
		MCDB	39.9	38.2	38.3	46.8	59.6	68.7	73.8	72.1	69.2	60.1	55.5	42.4
	2%	WB	34.4	33.7	34.3	39.5	50.8	59.5	65.3	65.4	61.9	53.1	46.9	38.2
		MCDB	34.9	34.7	35.4	41.6	56.1	65.8	71.2	70.5	66.1	55.5	48.2	38.9
	5%	WB	31.8	31.6	33.0	37.1	47.4	56.8	63.3	63.5	59.4	50.2	42.8	35.6
		MCDB	32.6	32.6	34.1	39.3	52.2	63.0	69.1	68.6	63.0	52.7	43.9	36.2
	10%	WB	29.2	29.2	31.4	35.1	44.3	54.3	61.3	61.8	57.0	47.2	39.7	33.1
		MCDB	30.1	30.3	32.7	36.9	48.8	60.6	67.1	66.8	60.6	49.5	40.9	33.8
Mean Daily Temperature Range	5% DB	MDBR	14.3	14.7	13.4	10.8	13.5	15.7	15.6	14.1	13.4	11.0	10.4	11.9
		MCDBR	13.5	13.4	12.4	13.1	21.6	22.5	20.6	18.4	16.6	15.6	13.5	11.4
		MCWBR	13.6	13.1	11.6	10.7	14.9	13.9	12.1	10.9	11.7	12.7	13.1	11.6
	5% WB	MCDBR	13.8	13.8	12.2	11.7	20.0	20.3	18.6	16.4	15.3	14.7	13.4	11.5
		MCWBR	13.9	13.8	11.8	10.0	14.8	13.9	12.4	11.0	12.4	12.9	13.3	11.8
Clear-Sky Solar Irradiance	taub		0.281	0.287	0.310	0.368	0.389	0.384	0.407	0.377	0.354	0.322	0.296	0.285
	taud		2.299	2.304	2.285	2.214	2.289	2.370	2.357	2.435	2.474	2.533	2.490	2.321
	Ebn at Noon		233	267	282	275	274	277	267	271	265	254	232	211
	Edh at Noon		20	27	33	40	39	37	37	32	28	22	17	17
All-Sky Solar Radiation	RadAvg		260	503	869	1250	1538	1602	1571	1332	982	585	312	203
	RadStd		35	78	124	168	149	143	127	89	74	48	29	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	+2.59	N/A	+0.67	+0.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page